$$\frac{2}{3}$$

$$\frac{2}{3}$$

$$\frac{2}{3}$$

$$-\frac{2}{3}$$

$$0$$

$$\frac{5}{6}$$

$$\frac{1}{4}$$

$$\frac{1}{2}$$

$$\frac{5}{6}$$

$$\frac{5}{7}$$

$$0$$

$$\infty$$

$$\infty$$

$$\infty$$

$$\frac{8}{27}$$

$$\frac{9}{4}$$

$$-\frac{8}{27}$$

$$\frac{2}{3}$$

$$\frac{2}{3}$$

$$\frac{2}{3}$$

$$-2$$

$$\frac{13717421}{219478738}$$

$$1$$

$$1267650600228229401496703205376$$

$$16$$

$$2y^3x^2\big|_{x=1,y=2} = 16$$

$$16$$

$$2y^3x^2$$

$$\frac{d}{dx}2y^3x^2 = 4y^3x$$

$$4y^3x$$

$$\frac{d}{dy}2y^3x^2 = 6y^2x^2$$

$$6y^2x^2$$

$$\frac{d}{dz}2y^3x^2 = 0$$

$$0$$

$$\int_1^2 2y^3x^2dx = \frac{14y^3}{3}$$

$$\frac{14y^3}{3}$$

$$\int_1^2 2y^3x^2dy = \frac{15x^2}{2}$$

$$\frac{15x^2}{2}$$

$$\int_1^2 2y^3x^2dz = 2y^3x^2$$

$$2y^3x^2$$

$$\frac{d}{dz}2y^3x^2 = 0$$

$$\frac{d}{dy}2y^3x^2 = 6y^2x^2$$

$$\frac{d}{dx}2y^3x^2 = 4y^3x$$

$$2y^3x^2 + 4y^3x + 6y^2x^2$$

$$\frac{d}{dz}2y^3x^2 = 0$$

$$\frac{d}{dy}2y^3x^2 = 6y^2x^2$$

$$\frac{d}{dx}2y^3x^2 = 4y^3x$$

$$2y^3x^2 + 4y^3x \geq 6x^2y^2$$

$$\frac{d}{dz}2y^3x^2 = 0$$

$$\frac{d}{dy}2y^3x^2 = 6y^2x^2$$

$$\frac{d}{dx}2y^3x^2 = 4y^3x$$

$$\frac{d}{dx}2y^3x^2 + 4y^3x \geq 6x^2y^2 = 4y^3 + 4y^3x \geq 12y^2x$$

$$4y^3 + 4y^3x \geq 12y^2x$$

$$\frac{5x}{7z} + \frac{4y}{7z}$$

$$2$$

$$2y$$

$$y + z$$

$$x$$

$$y + 2z$$

$$\frac{x^2 - y^2}{x - y}$$

$$\frac{x^2 + y^2 + 2xy}{x + y}$$

$$x + z$$

$$2y + 3z$$

$$3xy$$

$$\frac{x + y}{x + z}$$

$$y + \frac{z}{x}$$

$$1 + \frac{y}{x}$$

$$2x + 3y$$

$$1$$

$$0$$

$$\int_2^4 x^2 + z dx = 2z + \frac{56}{3}$$

$$2z + \frac{56}{3}$$

$$f(x,y) = \begin{cases} \frac{2x^{\frac{27}{10}}y^{\frac{13}{5}} + 11}{2y^{\frac{13}{5}}} & x >= y \\ \frac{2y^{\frac{27}{10}}x^{\frac{13}{5}} + 11}{2x^{\frac{13}{5}}} & x < y \\ 0 & otherwise \end{cases}$$